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Radio frequency ablation medical device

Abstract

A radio frequency ablation medical device includes a guide wire; a catheter configured to be movable to a lesion site of a tissue along the guide wire and having at least one electrode configured to generate heat according to an application of power; and a stent configured to be unfolded when protruding out of the catheter through an end portion of the catheter or recaptured in the end portion of the catheter when entering an inside of the catheter. The stent serves to transfer heat generated by the electrode to the lesion site in an unfolded state when getting out of the catheter.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8058771	12/2010	Giordano et al.	N/A	N/A
8253303	12/2011	Giordano et al.	N/A	N/A
8546996	12/2012	Messerly et al.	N/A	N/A
8704425	12/2013	Giordano et al.	N/A	N/A
8749116	12/2013	Messerly et al.	N/A	N/A
8779648	12/2013	Giordano et al.	N/A	N/A
8951248	12/2014	Messerly et al.	N/A	N/A
8956349	12/2014	Aldridge et al.	N/A	N/A
8986302	12/2014	Aldridge et al.	N/A	N/A
9039695	12/2014	Giordano et al.	N/A	N/A
9050093	12/2014	Aldridge et al.	N/A	N/A
9060775	12/2014	Wiener et al.	N/A	N/A
9060776	12/2014	Yates et al.	N/A	N/A
9072539	12/2014	Messerly et al.	N/A	N/A
9089360	12/2014	Messerly et al.	N/A	N/A
9168054	12/2014	Turner et al.	N/A	N/A
9504855	12/2015	Messerly et al.	N/A	N/A
9623237	12/2016	Turner et al.	N/A	N/A
9770353	12/2016	Shin et al.	N/A	N/A
9795808	12/2016	Messerly et al.	N/A	N/A
10022567	12/2017	Messerly et al.	N/A	N/A
10022568	12/2017	Messerly et al.	N/A	N/A
10201382	12/2018	Wiener et al.	N/A	N/A
10263171	12/2018	Wiener et al.	N/A	N/A
10265117	12/2018	Wiener et al.	N/A	N/A
10335614	12/2018	Messerly et al.	N/A	N/A

10441345	12/2018	Aldridge et al.	N/A	N/A
2009/0143777	12/2008	Pacey et al.	N/A	N/A
2010/0036405	12/2009	Giordano et al.	N/A	N/A
2011/0082486	12/2010	Messerly et al.	N/A	N/A
2011/0087212	12/2010	Aldridge et al.	N/A	N/A
2011/0087213	12/2010	Messerly et al.	N/A	N/A
2011/0087214	12/2010	Giordano et al.	N/A	N/A
2011/0087215	12/2010	Aldridge et al.	N/A	N/A
2011/0087216	12/2010	Aldridge et al.	N/A	N/A
2011/0087217	12/2010	Yates et al.	N/A	N/A
2011/0087256	12/2010	Wiener et al.	N/A	N/A
2012/0078139	12/2011	Aldridge et al.	N/A	N/A
2012/0123458	12/2011	Giordano et al.	N/A	N/A
2012/0265196	12/2011	Turner et al.	N/A	N/A
2012/0310262	12/2011	Messerly et al.	N/A	N/A
2012/0310263	12/2011	Messerly et al.	N/A	N/A
2012/0310264	12/2011	Messerly et al.	N/A	N/A
2013/0035706	12/2012	Giordano et al.	N/A	N/A
2013/0035707	12/2012	Giordano et al.	N/A	N/A
2015/0133927	12/2014	Shin et al.	N/A	N/A
2015/0182251	12/2014	Messerly et al.	N/A	N/A
2015/0182276	12/2014	Wiener et al.	N/A	N/A
2015/0182277	12/2014	Wiener et al.	N/A	N/A
2015/0196318	12/2014	Messerly et al.	N/A	N/A
2015/0327883	12/2014	Messerly et al.	N/A	N/A
2015/0328484	12/2014	Messerly et al.	N/A	N/A
2015/0340586	12/2014	Wiener et al.	N/A	N/A
2016/0089533	12/2015	Turner et al.	N/A	N/A
2017/0095267	12/2016	Messerly et al.	N/A	N/A
2017/0231694	12/2016	Mathur et al.	N/A	N/A
2018/0116706	12/2017	Wiener et al.	N/A	N/A
2018/0168714	12/2017	Wiener et al.	N/A	N/A
2019/0282292	12/2018	Wiener et al.	N/A	N/A
2020/0078609	12/2019	Messerly et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2559995	12/2012	EP	G01J 5/10
2013-541987	12/2012	JP	N/A
10-0472737	12/2004	KR	N/A
10-2010-0105365	12/2009	KR	N/A
10-2013-0140954	12/2012	KR	N/A

OTHER PUBLICATIONS

International Search Report issued Apr. 14, 2021 in PCT/KR2020/018704, filed on Dec. 18, 2020, 7 pages. cited by applicant

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The disclosure relates to a radio frequency ablation medical device.

RELATED ART

(2) Recently, various procedures have been performed using a method of applying high frequencies of various outputs to a human body at specific intervals.

(3) For example, various types of treatment, such as high frequency therapy for liver cancer, which applies high frequency heating to necrotic tumor cells in the liver, and high frequency therapy, which is used to treat muscle pain and joint pain by applying a radio frequency current to the human body to generate heat inside the human body, have been performed.

(4) As catheters for radio frequency ablation that are commercially available in clinical practice, there are Habib EndoHBP (EMcision, London, UK) and ELRA RFA catheters (Starmed, Goyang, Korea). The Habib catheter is an 8Fr bipolar catheter, and can be inserted into an endoscope along a guide wire in the form in which the Habib catheter has two 8 mm electrodes attached thereto. The ELRA catheter is a 7Fr bipolar catheter, and has four types: 11 mm, 18 mm, 22 mm, and 33 mm.

(5) However, for the treatment of malignant biliary stenosis including biliary tract cancer, the conventional radio frequency ablation (RFA) catheter used for radio frequency ablation (RFA) causes several problems.

(6) For example, when the radio frequency ablation catheter is applied to a curved biliary duct lesion, it may be difficult to uniformly transfer heat to tissues in the biliary tract, thermal ablation may be performed on an unwanted area through the radio frequency ablation catheter, and thermal ablation may not be sufficiently performed on a lesion site. In addition, in the case of the bipolar catheter having a diameter of 7Fr or 8Fr, since the catheter may not pass through the severe stenosis, treatment using the catheter may be difficult.

RELATED ART DOCUMENT

(7) Korean Patent Laid-Open Publication No. 10-2010-0105365 (published on Sep. 29, 2010)

SUMMARY

(8) Embodiments of the present disclosure provide a radio frequency ablation medical device capable of uniformly transferring heat to a tissue in a biliary tract and controlling an ablation range.

(9) In accordance with an embodiment of the present disclosure, there is provided a radio frequency ablation medical device including: a guide wire; a catheter configured to be movable to a lesion site of a tissue along the guide wire and having at least one electrode configured to generate heat according to an application of power; and a stent configured to be unfolded when protruding out of the catheter through an end portion of the catheter or recaptured in the end portion of the catheter when entering an inside of the catheter, the stent serving to transfer heat generated by the electrode to the lesion site in an unfolded state when getting out of the catheter.

(10) The catheter may include: a support tube configured to be movable along the guide wire; a peak tube supported by the support tube, wherein the peak tube is disposed in the stent so that the stent surrounds the peak tube; and a moving tube configured to accommodate at least a portion of the stent therein, the moving tube being movable along the support tube to adjust a length of the stent protruding out of the catheter.

(11) The radio frequency ablation medical device may further include: a thermal detection sensor for measuring an electrode temperature value of the electrode when the power is applied to the electrode.

(12) The radio frequency ablation medical device may further include a radio frequency generator

for applying power to the electrode, wherein the radio frequency generator may include: a power supply for applying power to the electrode; a temperature input unit configured to receive a specific temperature value of the electrode; a time input unit configured to receive a specific heating time during which the electrode is heated; an electrode temperature receiver for receiving the electrode temperature value of the electrode from the thermal detection sensor; and a temperature control unit configured to determine whether the electrode temperature value exceeds the specific temperature value or a heating time of the electrode exceeds the specific heating time and control the power supply to cut off the application of power to the electrode when it is determined that the electrode temperature value exceeds the specific temperature value or the heating time of the electrode exceeds the specific heating time.

(13) The radio frequency ablation medical device may further include: a radio frequency generator for applying power to the electrode, wherein the radio frequency generator controls the electrode temperature value of the electrode depending on an expansion length of the unfolded stent so that the electrode temperature value received from the thermal detection sensor satisfies a preset temperature compensation range.

(14) The temperature compensation range may be a temperature compensation range of the electrode for keeping a temperature of the unfolded stent constant when the expansion length is changed.

(15) The stent may be provided in a form of a cylindrical mesh which compresses and surrounds the peak tube by pressing of the moving tube in a space between the peak tube and the moving tube of the catheter.

(16) The radio frequency ablation medical device may further include: a pusher connected to the moving tube, wherein the stent is unfolded from the peak tube to be in close contact with the lesion site when the pressing of the moving tube is released according to a movement of the moving tube in one direction in response to a pull operation of the pusher, and the stent is recaptured into the catheter by the pressing of the moving tube according to a movement of the moving tube in the other direction in response to a push operation of the pusher.

(17) The radio frequency ablation medical device may further include: a radio frequency generator for applying power to the electrode, wherein the radio frequency generator further includes a temperature compensation calculator that calculates a temperature compensation range of the electrode for keeping a temperature of the stent constant depending on an expansion length of the stent.

(18) After determining whether the electrode temperature satisfies the temperature compensation range, the temperature controller may control the temperature of the electrode so that the electrode temperature satisfies the temperature compensation range.

(19) The radio frequency ablation medical device may further include: a thermal image sensor for measuring a temperature for each length section of the stent.

(20) The radio frequency ablation medical device may further include: a radio frequency generator that applies power to the electrode, wherein the radio frequency generator receives temperature information for each length section of the stent from the thermal image sensor, and controls the temperature of the electrode so that the temperature for each length section of the stent satisfies a preset temperature section range.

(21) According to the embodiments of the present disclosure, it is possible to perform radio frequency ablation using a stent widely used for nonvascular stenosis disease, uniformly transfer heat to a lesion site of a tissue in a biliary tract, and control an ablation range for the lesion site.

(22) In addition, the embodiments of the present disclosure can be applied to various non-vascular organs that require local thermal ablation, including a biliary tract.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a configuration diagram illustrating a radio frequency ablation medical device according to an embodiment of the present disclosure.

(2) FIG. 2 is an enlarged view of part A of FIG. 1.

(3) FIG. 3 is a state diagram illustrating a state in which a stent of the radio frequency ablation medical device according to the embodiment of the present disclosure is unfolded at a lesion site.

(4) FIG. 4 is a block diagram illustrating a control flow of a radio frequency generator of the radio frequency ablation medical device according to the embodiment of the present disclosure.

(5) FIG. 5 is a block diagram illustrating a control flow of a radio frequency generator of a radio frequency ablation medical device according to a modification of the present disclosure.

(6) FIG. 6 is a block diagram illustrating a control flow of a radio frequency generator of a radio frequency ablation medical device according to another embodiment of the present disclosure.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

(7) Hereinafter, a configuration and operation according to an embodiment of the present disclosure will be described in detail with reference to the accompanying drawings. The following description is one of many claimable aspects of the disclosure and may form part of the detailed description of the disclosure. However, in describing the present disclosure, detailed descriptions of known configurations or functions may be omitted to clarify the present disclosure.

(8) The present disclosure may make various changes and include various embodiments, and specific embodiments will be illustrated in the drawings and described in the detailed description. However, the present disclosure is not limited to a specific embodiment, and it should be understood that the present disclosure includes all changes, equivalents, or substitutes included in the spirit and scope thereof.

(9) Terms including ordinal numbers, such as first and second, may be used for describing various elements, but the corresponding elements are not limited by these terms. These terms are only used for the purpose of distinguishing one element from another element. When an element is referred to as being 'connected' to or 'accessed' by another element, it should be understood that the element may be directly connected to or accessed by the other element, but that other elements may exist in the middle. The terms used in the present disclosure are only used for describing specific embodiments, and are not intended to limit the present disclosure. Singular expressions include plural expressions unless the context clearly indicates otherwise.

(10) FIG. 1 is a configuration diagram illustrating a radio frequency ablation medical device according to an embodiment of the present disclosure, FIG. 2 is an enlarged view of part A of FIG. 1, FIG. 3 is a state diagram illustrating a state in which a stent of the radio frequency ablation medical device according to the embodiment of the present disclosure is unfolded at a lesion site, and FIG. 4 is a block diagram illustrating a control flow of a radio frequency generator of the radio frequency ablation medical device according to the embodiment of the present disclosure.

(11) As illustrated in FIGS. 1 to 4, a radio frequency ablation medical device **10** according to the embodiment of the present disclosure may include a guide wire **100**, a catheter **200**, a stent **300**, a thermal detection sensor **700**, and a radio frequency generator **400**.

(12) Specifically, the guide wire **100** may be provided in the form of a wire for guiding a catheter **200** to a lesion site L generated in a patient's tissue (e.g., tissue in a biliary tract). The guide wire **100** may be mounted in a manner that penetrates through the catheter **200**, and the catheter **200** may move along the guide wire **100**.

(13) The catheter **200** may be moved to the lesion site L of the tissue along the guide wire **100**. In a state where the catheter **200** is positioned at the lesion site L of the tissue, thermal ablation may be performed on a lesion B of the tissue using heat provided through the stent **300**. An electrode **600** for providing heat to the stent **300** may be provided in a front end portion of the catheter **200**.

(14) The catheter **200** may include a support tube **210**, a peak tube **220**, a hub **230**, a moving tube

240, and a pusher **250**. The support tube **210** may be provided in the form of a hollow tube movable along the guide wire **100**. A front end of the support tube **210** may be coupled to the peak tube **220**, and a rear end of the support tube **210** may be coupled to the hub **230**.

(15) The peak tube **220** may be provided in the form of a hollow tube movable along the guide wire **100**. A stent **300** may be mounted on the circumference of the peak tube **220**. A conical guide tip **260** may be coupled to the front end of the peak tube **220**, and a rear end of the peak tube **220** may be coupled to the support tube **210**.

(16) The peak tube **220** may be provided with an electrode **600**. The electrode **600** may be electrically connected to the radio frequency generator **400**. The electrode **600** may receive power required for heat generation from the radio frequency generator **400**. The electrode **600** may generate heat and provide the heat to the stent **300**. The electrode temperature generated from the electrode **600** may satisfy the range of 70° C. to 100° C.

(17) In the present embodiment, at least one or more (for example, 3 to 5) electrodes **600** are disposed spaced apart from the front end portion of the catheter **200** (for example, the front end portion of the peak tube **220**), but is not limited thereto, and in addition to the electrode, various types of heating elements capable of heating the stent **300** through the heat generation may be provided in the front end portion of the catheter **200**. As the heating element, a hot wire capable of generating heat may be used.

(18) The hub **230** may be coupled to the rear end of the support tube **210**. A support tube **241** may penetrate through the hub **230** to be connected to the moving tube **240**. The support tube **241** may be provided in the form of a hollow tube connecting the moving tube **240** and the pusher **250**. The guide wire **100** may penetrate through the support tube **241**. When a user pulls or pushes the pusher **250**, the support tube **241** may transmit an operating force of the pusher **250** to the moving tube **240**.

(19) The moving tube **240** may be provided in the form of a tube in which at least a portion (e.g., front end portion) thereof surrounds the stent **300**. The moving tube **240** may be connected to the pusher **250** via the support tube **241**. Accordingly, during the pull or push operation of the pusher **250**, the moving tube **240** may move backward or forward along the support tube **210**. The moving tube **240** may be made of a material capable of blocking heat conduction.

(20) In particular, when the moving tube **240** is moved in a rearward direction (a direction in which the pusher is pulled) along the support tube **210**, the stent **300** may be unfolded out from the moving tube **240**. In this case, an expansion length E of the stent **300** in which the stent **300** is unfolded may be controlled in proportion to the moving distance of the moving tube **240** moving in the rear direction. Further, when the moving tube **240** is moved in a forward direction (a direction in which the pusher is pushed) along the support tube **210**, the stent **300** may be recaptured by the moving tube **240**.

(21) The stent **300** may be provided in the form of a cylindrical mesh. The stent **300** may be unfolded to be controllable in length in the end portion of the catheter **200** or may be recaptured into the end portion of the catheter **200**.

(22) For example, when the stent **300** may be pressed by the moving tube **240**, the stent **300** is located in a space between the peak tube **220** and the moving tube **240** of the catheter **200**, and therefore, may be maintained in a compressed state while surrounding the peak tube **220**. When the pressure of the moving tube **240** with respect to the stent **300** is released, the stent **300** may be restored to the original form of the cylindrical mesh while being expanded. When the stent **300** is unfolded from the end portion of the catheter **200**, the stent **300** may provide heat generated from the electrode **600** to the lesion site L.

(23) Accordingly, during the pull operation of the pusher **250**, when the pressure of the moving tube **240** with respect to the stent **300** is released according to the rearward movement of the moving tube **240**, the stent **300** may be unfolded from the front end portion of the peak tube **220** so as to be in close contact with the lesion site L. During the push operation of the pusher **250**, when

the moving tube **240** is pressed against the stent **300** according to the forward movement of the moving tube **240**, the stent **300** may be recaptured into the space between the moving tube **240** and the support tube **210** of the catheter **200**.

(24) The thermal detection sensor **700** may include a temperature sensor capable of measuring an electrode temperature generated from the electrode **600**. Since the thermal detection sensor **700** is installed in contact with the electrode **600** or installed adjacent to the electrode **600**, it is possible to measure the electrode temperature generated from the electrode **600**. The information on the electrode temperature value of the electrode **600** measured by the thermal detection sensor **700** may be applied to the radio frequency generator **400**.

(25) The radio frequency generator **400** may be electrically connected to the electrode **600** to apply power to the electrode **600**. The radio frequency generator **400** may include a power supply unit **410**, a temperature input unit **420**, a time input unit **430**, an electrode temperature receiver **440**, and a temperature controller **450**.

(26) The power supply unit **410** may receive power from an external power supply, and provide at least a portion of the received power to the electrode **600**. The temperature input unit **420** may receive a specific temperature of the electrode **600** from the user. Here, the specific temperature may be understood as a maximum temperature of the electrode **600** at which the application of power to the electrode **600** is permitted.

(27) The time input unit **430** may receive a specific time at which the heating of the electrode **600** from the user is performed. Here, the specific time may be understood as a maximum time during which the application of power to the electrode **600** is permitted. The electrode temperature receiver **440** may receive an electrode temperature value of the electrode **600** measured from the thermal detection sensor **700**.

(28) When the specific temperature provided through the temperature input unit **420** exceeds the electrode temperature value applied from the electrode temperature receiver **440**, the temperature controller **450** may control the power supply unit **410** so that the application of power to the electrode **600** is automatically cut off. For example, when the specific temperature provided through the temperature input unit **420** is 80° C. and the electrode temperature value applied from the electrode temperature receiver **440** is 75° C., the temperature controller **450** may apply power to the electrode **600** until the electrode temperature value reaches 80° C., and when the electrode temperature value exceeds 80° C., the temperature controller **450** may cut off the application of power to the electrode **600**. In this way, when the application of power to the electrode **600** is controlled based on a specific temperature, damage to an organ due to an excessive increase in temperature may be prevented in advance.

(29) In addition, when the heating time of the electrode **600** exceeds a specific time provided through the time input unit **430**, the temperature controller **450** may control the power supply unit **410** to automatically cut off the application of power to the electrode **600**. For example, when the specific time provided through the time input unit **430** is 1 minute, the temperature controller **450** may apply power to the electrode **600** until the heating time of the electrode **600** becomes 1 minute, and when the heating time of the electrode **600** exceeds 1 minute, the temperature controller **450** may cut off the application of power to the electrode **600**.

(30) Meanwhile, when power is supplied to the electrode **600** by the radio frequency generator **400**, even if the mass of the stent **300** itself to which the heat generated from the electrode **600** is transferred is constant, since the expansion (unfolded) portion of the stent **300** is in contact with the patient's lesion site L, considering a patient's body having a larger specific heat than that of the stent **300**, when the expansion length E of the stent **300** is long, the amount of heat required to reach a predetermined target temperature (e.g., a specific temperature) is required more than when the expansion length E of the stent **300** is relatively small. Therefore, it is necessary to correct the relationship between the expansion length E of the stent **300** and the amount of heat generated from the electrode **600**.

(31) Accordingly, the radio frequency generator **400** may control the temperature of the electrode **600** in accordance with the change in the expansion length E at which the stent **300** is unfolded, so that the electrode temperature applied from the thermal detection sensor **700** satisfies the preset temperature compensation range.

(32) Here, the temperature compensation range may be the temperature range of the electrode **600** set to keep the temperature of the stent **300** constant even if the expansion length E of the stent **300** changes. The temperature compensation range may be preset through data calculated through a temperature relationship of the electrode **600** according to the change in the expansion length E.

(33) For example, if the standard temperature of the electrode **600** required for thermal ablation of the lesion site L is 80° C. when the expansion length E of the stent **300** is 20 mm, the temperature compensation range of the electrode **600** required for thermal ablation of the lesion site L may be preset in the range of 80.5° C. to 81.5° C. when the expansion length (E) of the stent **300** is 30 mm through the data on the temperature relationship of the electrode **600** according to the change of the expansion length E.

(34) After all, for the thermal ablation of the lesion site L, when the expansion length E of the stent **300** is adjusted from 20 mm to 30 mm, the radio frequency generator **400** may heat or lower the temperature of the electrode **600** based on the expansion length E adjusted to 30 mm so that the electrode temperature applied from the thermal detection sensor **700** satisfies the preset temperature compensation range of 80.5° C. to 81.5° C.

(35) FIG. 5 is a block diagram illustrating a control flow of a radio frequency generator of a radio frequency ablation medical device according to a modification of the present disclosure.

(36) As illustrated in FIG. 5, according to a modification of the present disclosure, the radio frequency generator **400** may include a power supply unit **410**, a temperature input unit **420**, a time input unit **430**, an electrode temperature receiver **440**, a temperature compensation calculator **460**, and a temperature controller **450**. Here, the configurations other than the temperature controller **450** and the temperature compensation calculator **460** correspond to the configurations described above, and therefore, a detailed description thereof will be omitted.

(37) The temperature compensation calculator **460** may calculate the temperature compensation range of the electrode **600** through data on the temperature relationship of the electrode **600** according to the change in the expansion length E in order to keep the temperature of the stent **300** constant according to the change of the expansion length E.

(38) After determining whether the electrode temperature satisfies the temperature compensation range calculated by the temperature compensation calculator **460**, the temperature controller **450** may control the temperature of the electrode **600** so that the electrode temperature satisfies the temperature compensation range.

(39) For example, when the expansion length E of the stent **300** is adjusted from 20 mm to 30 mm, the radio frequency generator **400** may heat or lower the temperature of the electrode **600** based on the expansion length E adjusted to 30 mm so that the electrode temperature applied from the thermal detection sensor **700** satisfies the temperature compensation range of 80.5° C. to 81.5° C. calculated by the temperature compensation calculator **460**.

(40) FIG. 6 is a block diagram illustrating a control flow of a radio frequency generator of a radio frequency ablation medical device according to another embodiment of the present disclosure.

(41) As illustrated in FIG. 6, the radio frequency ablation medical device **10** according to the embodiment of the present disclosure may further include a thermal image sensor **800**.

(42) The thermal image sensor **800** may measure the temperature for each length section of the stent **300**. Here, the temperature for each length section may be understood as an average value of temperatures measured in a plurality of sections arbitrarily dividing the stent **300** along the longitudinal direction. The information on the temperature for each length section of the stent **300** measured through the thermal image sensor **800** may be applied to the radio frequency generator **400**.

(43) The radio frequency generator **400** may control the temperature of the electrode **600** so that the temperature for each length section of the stent **300** applied from the thermal image sensor **800** satisfies a preset temperature section range.

(44) For example, when the preset temperature section ranges from 85° C. to 86° C., the temperature for each length section of the stent **300** obtained from the thermal image sensor **800** is 83° C., the radio frequency generator **400** may heat the temperature of the electrode **600** so that the temperature for each length section satisfies the preset temperature section ranging from 85° C. to 86° C.

(45) Through this, the temperature of the stent required for thermal ablation to the lesion site L of the tissue may be kept constant, and as a result, uniform heat transfer to the lesion site L may be possible.

(46) As described above, the present disclosure has excellent advantages such as performing that radio frequency ablation using a stent widely used for nonvascular stenosis disease, uniformly transferring heat to the lesion site of the tissue in the biliary tract, controlling the ablation range for the lesion site, and being applied to various non-vascular organs that require local thermal ablation, including the biliary tract.

(47) The embodiments described above are merely some examples illustrating the present technical spirit, and the technical scope of the present disclosure is not limited by the embodiments. It will be understood by those skilled in the art that various changes, modifications and substitutions can be made without departing from the technical scope of the present disclosure, and all such implementations should be considered to fall within the scope of the present technical spirit.

(National R&D project supporting the present invention) (Project number) 1711112394 (Project unique number) 2020R1A2C2003604 (Ministry name) Ministry of Science and Technology Information and Communication (Name of project management (specialized) institution): Specialized institution: National Research Foundation of Korea (Research project name) Personal basic research project (middle-level research) (Research project name) Development of next-generation radio frequency ablation medical device for treatment of malignant bile duct stenosis (Contribution Rate) 1/1 (Name of organization performing task) Ulsan University Industry-Academic Cooperation Foundation (Research Period) 2020.03.01 to 2023.02.28

Claims

1. A radio frequency ablation medical device comprising: a guide wire; a catheter configured to be movable to a lesion site of a tissue along the guide wire and having at least one electrode configured to generate heat according to an application of power; a stent configured to be unfolded when protruding out of the catheter through an end portion of the catheter or recaptured in the end portion of the catheter when entering an inside of the catheter, the stent serving to transfer heat generated by the electrode to the lesion site in an unfolded state when getting out of the catheter; a thermal detection sensor for measuring an electrode temperature value of the electrode when the power is applied to the electrode; and a radio frequency generator for applying power to the electrode, wherein the radio frequency generator controls the electrode temperature value of the electrode depending on an expansion length of the unfolded stent so that the electrode temperature value received from the thermal detection sensor satisfies a preset temperature compensation range.
2. The radio frequency ablation medical device of claim 1, wherein the catheter includes: a support tube configured to be movable along the guide wire; a peak tube supported by the support tube, wherein the peak tube is disposed in the stent so that the stent surrounds the peak tube; and a moving tube configured to accommodate at least a portion of the stent therein, the moving tube being movable along the support tube to adjust a length of the stent protruding out of the catheter.
3. The radio frequency ablation medical device of claim 1, wherein the radio frequency generator includes: a power supply for applying power to the electrode; a temperature input unit configured

to receive a specific temperature value of the electrode; a time input unit configured to receive a specific heating time during which the electrode is heated; an electrode temperature receiver for receiving the electrode temperature value of the electrode from the thermal detection sensor; and a temperature control unit configured to determine whether the electrode temperature value exceeds the specific temperature value or a heating time of the electrode exceeds the specific heating time and control the power supply to cut off the application of power to the electrode when it is determined that the electrode temperature value exceeds the specific temperature value or the heating time of the electrode exceeds the specific heating time.

4. The radio frequency ablation medical device of claim 1, wherein the temperature compensation range is a temperature compensation range of the electrode for keeping a temperature of the unfolded stent constant when the expansion length is changed.

5. The radio frequency ablation medical device of claim 1, wherein the stent is provided in a form of a cylindrical mesh which compresses and surrounds the peak tube by pressing of the moving tube in a space between the peak tube and the moving tube of the catheter.

6. The radio frequency ablation medical device of claim 5, further comprising: a pusher connected to the moving tube, wherein the stent is unfolded from the peak tube to be in close contact with the lesion site when the pressing of the moving tube is released according to a movement of the moving tube in one direction in response to a pull operation of the pusher, and the stent is recaptured into the catheter by the pressing of the moving tube according to a movement of the moving tube in the other direction in response to a push operation of the pusher.

7. A radio frequency ablation medical device comprising: a guide wire; a catheter configured to be movable to a lesion site of a tissue along the guide wire and having at least one electrode configured to generate heat according to an application of power; a stent configured to be unfolded when protruding out of the catheter through an end portion of the catheter or recaptured in the end portion of the catheter when entering an inside of the catheter, the stent serving to transfer heat generated by the electrode to the lesion site in an unfolded state when getting out of the catheter; a thermal detection sensor for measuring an electrode temperature value of the electrode when the power is applied to the electrode; and a radio frequency generator for applying power to the electrode, wherein the radio frequency generator further includes a temperature compensation calculator that calculates a temperature compensation range of the electrode for keeping a temperature of the stent constant depending on an expansion length of the stent.

8. The radio frequency ablation medical device of claim 7, wherein, after determining whether the electrode temperature satisfies the temperature compensation range, the temperature controller controls the temperature of the electrode so that the electrode temperature satisfies the temperature compensation range.

9. A radio frequency ablation medical device comprising: a guide wire; a catheter configured to be movable to a lesion site of a tissue along the guide wire and having at least one electrode configured to generate heat according to an application of power; a stent configured to be unfolded when protruding out of the catheter through an end portion of the catheter or recaptured in the end portion of the catheter when entering an inside of the catheter, the stent serving to transfer heat generated by the electrode to the lesion site in an unfolded state when getting out of the catheter; a thermal image sensor for measuring a temperature for each length section of the stent; and a radio frequency generator that applies power to the electrode, wherein the radio frequency generator receives temperature information for each length section of the stent from the thermal image sensor, and controls the temperature of the electrode so that the temperature for each length section of the stent satisfies a preset temperature section range.
